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ON ARTIFICIAL INTELLIGENCE AND THEORY CONSTRUCTION IN SOCIOLOGY

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1. INTRODUCTION

Among the behavioral sciences sociology is less influenced by the developments in Artificial Intelligence and Cognitive Science than the others. For sure, computer simulations have been used by sociologists for a long time: there are computer models of voting behavior, social mobility, decision-making in organizations, urban decay and many other social processes. But most computer simulation is not Artificial Science. Cognitive science is not the same as Artificial Intelligence, although the two fields are closely linked in the minds and procedures of many researchers. The recent history of cognitive science by Howard Gardner (Gardner, 1985) discusses the participation of psychologists, linguists, philosophers, anthropologists and neuroscientists in "the cognitive revolution", but sociologists are not mentioned. This is not the place to investigate the detailed reasons for the reluctance of sociologists to participate in one of the most fruitful movements in contemporary behavioral science. One reason may be that sociologists are, since Durkheim, preoccupied with finding for themselves a *concrete* set of phenomena that is "uniquely sociological". This makes them see certain other sets of concrete phenomena as already claimed for other disciplines, and therefore not the proper study for sociologists. For example, Granovetter has pointed out that sociologists in our time rarely study certain economic phenomena (Granovetter, 1985). The classics, e.g. Max Weber made major contributions to the study of economic life, but contemporary sociologists seem to regard markets, business decisions and transactions and the organization of industries as the province of microeconomics. This is simply confusing *concrete* sociological *subject matter* with the sociological *perspective*. A second reason may be sociologists' eagerness to define structure as the proper sociological concern and, hence, to downplay what happens "inside" the actor. But we also know that sociology does have, in the symbolic interactionisms, a considerable intellectual tradition that is, at least in part, cognitively oriented.

It is time for sociology to break its intellectual isolation and participate in the cognitivist rethinking of human action, and to avail itself of theoretical ideas, techniques and tools that have been developed in AI and cognitive science. This does not mean

that sociologists should become cognitive psychologists or broad practitioners of AI. Sociologists will, in fact, find much to intellectually disagree with in cognitive science, and an infusion of sociological thinking might help reinforce some of the criticisms of individual-centrism in cognitivist research, already advanced by cultural anthropologists like Clifford Geertz (Gardner, 243f., 355–359). Sociologists will also very likely also find that many of the topics studied in AI seem far removed from their own professional concerns. However, one should be cautious before one acts on such impression to write off substantive AI concerns as uninteresting for sociologists; problem areas that seem far removed from sociology might still in important respects be *isomorphic* with sociological ones. Such isomorphisms, often in the history of scientific fields revealed through appropriate formalisms, may prove to be fruitful departures for theory development.

How does cognitive science differ from the traditional cognitive orientations in psychology, phenomenology and social psychology, including the interactionist “sociological social psychology” that stems from G.H. Mead? The difference is a major one, but even so a matter of degree. Cognitive scientists attempt to specify *and* model psychological mechanisms in detail. Mechanisms with related functions are organized into functional modules. The ordering and interactions of the modules is seen to be of great importance in theory construction. Good illustrations of the strategy can be found, for example, in Marr’s work on vision (Marr, 1982), and in the work of Kosslyn and his coworkers (e.g. Pinker 1985, and references therein). For sociologists it may be of some interest that the thinking is *functionalist*, approximately in the technical sense the term was used by Talcott Parsons (e.g. Parsons, 1945, not to be confused with “functional explanation”). The first major problem in this type of theory construction will, of course, be the specification of the modules. In some cases there are natural solutions: in the model of a political system sketched later in this paper various major roles in the system obviously make up modules. A second and more difficult problem is the exact specification of each module’s mechanisms. When modeling a social system it is natural to think of the modules as receiving inputs from other modules, sending outputs to some of them, but each doing its own internal processing of inputs and its decision-making hidden from view of the other modules. A prominent concept for cognitive scientists is “mental representation”. Using abstract structures “in the mind” we create representations of pictures, persons, events, sentences, sounds etc. Sociological modules that correspond to individual or collective social actors or sets of roles should clearly contain carefully specified mental representations for processing their inputs. The use of mental representations as major components of models would set cognitive science and AI inspired sociological theory from ordinary computer simulations, and would also bring formal sociological theorizing closer to how we normally think about actors. Fortunately, some of the best work in cognitive science and AI has been devoted precisely to the modeling of mental representations. There is a good deal of useful work to use or draw on for suggestions, e.g. the formalism of primitive acts invented by Schank (see e.g. Schank and Abelson, 1977), and the discussion of “cause”, “can”, “knowledge” and other concepts by McCarthy and Hayes (McCarthy and Hayes, 1969).

In this paper I want to discuss how some concepts and techniques from AI and cognitive science can, if used properly, contribute to theory development in sociology.

I shall discuss two topics: conceptual analysis, and the modeling of complex, interactive social systems. Both are underdeveloped areas in sociological theory. In the natural sciences they have, of course, received a great deal of attention. One reason for this is that many, if not most, sociologists subscribe to empiricist philosophies of science or methodologies that define data collection and data analysis as the major respectable scientific activities, while conceptual work and meticulous theory construction are seen as preliminaries or "tag-ons". This attitude holds back the discipline. Deeper conceptual analyses than the present largely verbal theories allow, will uncover theoretical and empirical problems for investigation. In other words, there is knowledge to be gained through conceptual analysis, not merely better "nominal definitions" (see also Wang, 1985). Complex interactive systems can only be understood through modeling, for verbal formulations do not allow us to get detailed knowledge about trajectories, or conditions of system stability or instability.

As illustrations I will use the role concept, and the problem of how to model a centralized, coercive political system. I have Mexico in mind, because I have previously done some work on Mexican politics, but the class of relevant systems is quite large, and my aim is to make some quite general points about theory construction in sociology. A verbal description of the Mexican system has been previously published by Manuel L. Carlos and myself (Carlos and Anderson 1981).

The sociologist is likely to find the AI field very technical and complex, and it is. The programming languages, techniques of representation and tools are evolving rapidly. However, I believe that at present, sociologists have more to learn from certain basic theoretical ideas than from particular AI implementations. Some sociologists might want to go into AI, and then they will have to retrain themselves for a new profession. The sociologist who wants to learn from AI in order to further the sociological work, needs to study concepts and paradigms in AI and cognitive science. In the brief and programmatic discussion in this paper I draw mostly on work by Minsky (Minsky, 1975), McCarthy and Hayes, (McCarthy and Hayes, 1969; McCarthy 1979) and Newell (Newell, 1982). Their work enjoys high esteem in the AI and cognitive science communities. They also represent quite different approaches to the fields. But fortunately the working sociologist does not need to take any position on the points of dispute between AI "schools".

2. THE ROLE CONCEPT

Since "role" is one of sociology's central concepts, one might expect that by now we would have many detailed descriptions and analyses of particular role-structures. This is not the case: role analyses tend to be sketchy and illustrate at best. One of the most detailed descriptions is still that given by Parsons of the patient role in *The Social System*. Why this is so is worth thinking about. Pre-Chomskyan linguists created detailed structural descriptions of many languages and cultural anthropologists have produced detailed ethnographies. Maybe, since sociologists have for the most part studied North-American or European institutions, they have tended to regard meticulous description as pedantry. The temptation is to believe that the readers are able to fill in the details themselves, once one has, so to speak, evoked the context.

This practise should be abandoned. I suggest that we will learn much about familiar roles through pondering detailed descriptions, just as linguists have learned about grammar by thinking about problematic sentences like "They are flying machines".

A notable exception is the work of Fararo and Skvoretz. They start with a case we all think we know about, namely the transactions between the waitress and customers in an ordinary restaurant, and proceed to spell out the institutionalized rules for such interactions. What they have is not a simple system, and one can very easily think of non-trivial additions that need to be made, e.g. meta-rule for how the basic production rules can be ordered, prototypes of regular and "problem" customers and so on. But methodologically, Fararo and Skvoretz are on the right track, for we must learn how to model and analyze role-systems as simple as this before we can turn to more complex forms, e.g. systems that involve "phase movements" and system levels. (For details the reader should turn to the papers by Fararo and Skvoretz; e.g. Fararo and Skvoretz, 1985, a and b, and Skvoretz, 1985).

Minsky's concept "frame" was developed for the detailed description of complexly ordered domains that contain a variety of objects, scenes, situations, rules, behavioral regularities, and events. (One of Minsky's examples is: What does one need to know in order to attend a child's birthday party?) The concept "role" as used in sociology refers to such domains. When we speak about the role of the teacher or the physician we *intend* (in the technical logical sense; see Husserl (1970), for a classical analysis, of intentionality, and Searle, (1983), for a modern discussion) a very complex and abstract object. Role behaviors are of course rather concrete, but roles are not role behaviors any more than the different percepts of a house together make up the house. We can loosely say that roles consists of rules, prototypes of scenes, actors and actions, practises, sanctions and relations e.g. time-relations and hierarchical orderings between these different objects and events. We can also say, maybe more precisely, that the conditions of satisfaction of statements about roles are defined in terms of rules, scenes, actors, actions etc. (For a discussion of the concept of "satisfaction" see Searle, 1983).

Minsky's frame conceptualization has been refined and formalized in AI research (see for example, Fikes and Ehler 1985, and references therein; Winston, 1984, Chapter 8). For sociologists the study of frame implementations in AI are useful in at least two different ways: First, one gets insights into how complex domains can be analyzed, comprehensively and systematically, *in practice*. Specialized frame languages have been developed that provide precise notations for domains of complex objects, scenes, production rules and so on. Having a suitable notation is, of course, in itself an enormous advantage in any analytical work. AI researchers analyze domains in order to implement the resulting representation in a computer program; they are therefore forced to strive for a very high degree of descriptive completeness. Sociologists can benefit from learning enough notation and acquiring the discipline of descriptive comprehensiveness. This is a *heuristic* use of the AI frame concept. Second, once a role structure has been defined and analyzed, there are substantive sociological problems that can be investigated through the use of the formal techniques that some AI representations provide. For example, sociological role analyses and students of rule systems in general are familiar with role-conflict and norm-conflict. In a classical paper Robert K. Merton studied the role-set, (Merton, 1957,

425–435). At issue are the mechanisms and strategies by which social actors attempt to manage (“articulate”) contradictory role expectations. I propose that a formal analysis of role-sets along the lines suggested here would lead to progress in the study of Merton’s problem. A formalization would express the role-expectations as production rules in terms of the first-order predicate calculus. Logic programs can locate *precisely* any contradictions between statements. When contradictions have been precisely identified the researcher can go back to informants with further questions about how conflicts are dealt with. This *substantive* use of the AI representation may lead to an interaction between theory construction and empirical research. Such interactions appear to be routine in the development of expert programs in AI.

3. COMPLEX POLITICAL SYSTEMS

The scientific study of political systems needs to pay less attention to gathering and interpreting more or less striking and interesting particular events, and more to spelling out in some detail the *abstract principles* that rule the systems and account for their continuity and reproduction in the middle of exogenously and endogenously generated change. This is in line with the perspective taken for granted in the natural sciences, and also in the most exciting research programs in contemporary humanities, e.g. the French *Annales* school of historiography (e.g. Braudel, 1980). Aristotle, the founder of scientific methodology, knew very well that knowledge can only be had about *classes* and *kinds* of phenomena, but his insight got lost in some social science fields. I suggest that we should look at the political system that exists at a particular in time and place as an instantiation of an underlying abstract structure. Sociologists who adopt this way of thinking might still make a science of their discipline, by taking part in the quiet revolution in the human sciences that started with Chomsky’s announcement almost thirty years ago of his research program in linguistics, (Chomsky, 1957, Lees, 1957).

A political system may for modeling purposes be thought of as an abstract structure, composed of idealized actors and the mutliplex relations between these. The structure is *abstract*, because we conceive of the actors only as incumbents of political *roles*, i.e. we are not modeling individuals with many interests or individual tastes and diverse motives. The structure is *idealized*, because, in order to make the model feasible, we ignore subtypes of political roles. In other words, while the model attempts to capture the basic features of a political system there remains much variation, e.g. between regional or other local systems, that it for the time ignores. The model will most likely be built incrementally, and some of the variation left out in early versions will be included in later ones. (For useful remarks on abstract and concrete models, see also Bailey, 1983.)

A partial list of the prototypical actors (modules) in a model of the Mexican polity would be: The President of the Republic and his internal security advisor, the army command, the leader of the ruling party on the local level, the skilled urban worker, the minister of economic planning, the peasant proprietor, the landless peasant, the urban slum dweller, the urban middle class professional. Actors pass messages to other actors; there may take the form of threats, appeals, positive or negative

sanctions, expressions of solidarity or antagonism. Actors also engage in *social comparisons* with other relevant actors, in terms of their standing on various rank dimensions, e.g. income, power, and possession of status objects. Such comparisons are believed to be important determinants of the sense of justice or relative deprivation. Actors can also form coalitions with other actors on the basis of rank comparisons, (see Zelditch and Anderson, 1967).

Each actor role has several *action options*. Thus, the President can pursue policies favorable to the interests of some of the other actors and unfavorable to those of others or order repression of some actor or coalition. Urban workers can choose to support the ruling party join an opposition coalition, go on strike etc.

The system exists in an *environment* in which certain law-like regularities obtain. These are primarily economic laws. They define, for example, the relationship between the size of the foreign debt and the resources available for job creation and improvements in the standard of living, the prices of the system's exports and imports and demographic trends. (For relevant research see e.g. Sachs, 1985 and references therein.)

The actors are assumed to have *purposes*, i.e. interests that they attempt to maximize or satisfice, and also *world-views*, i.e. cognitive structures, composed of concepts and beliefs, that are used to represent the various *inputs* received from both other actors and the system's environment. For each actor there are also production *rules* that connect the interpretations of the input from other actors and the events in the environment with the available action options. The actor's cognitive structures, in combination with the inputs, produce beliefs about causes, about what other actors did, could have done and will do, and what the actor himself can do.

The sociologist who seriously wants to model the system will have to overcome many empirical difficulties, because of the research literature about the system is not likely to contain all the desired information. Informants will in many cases be able to supply desired information; as with expert systems in AI, social system modeling may require a close interaction between the construction of formal representations and detailed fieldwork with qualified informants.

The description given here is obviously a mere crude sketch of a model of a political system. The details do not matter for the points about sociological theory construction that I want to make here. Social systems are composed of purposive and intelligent actors. "Intelligent" here means that they possess concepts, knowledge and beliefs, that they acquire knowledge and beliefs as part of the social process and that they draw inferences, make plans and choose actions on the basis of their knowledge, beliefs and purposes. To make progress, sociological theorists will have to study *and* model the *details* of these aspects of actors' behavior. Not to do this reduces most explanations of actors' behavior to hand-waving, and makes prediction a matter of guesswork or opinion. The sociologist who seriously wants to model the cognitive structures of social actors has two literatures to draw from formal means of representation: the study of belief and other propositional attitudes in formal logical and semantics (e.g. Hintikka 1962, Barwise and Perry, 1984), and the work on knowledge representations and related topics in AI. I am proposing that sociologists should take seriously the common sense knowledge that they (and laymen) already have about social roles and actors, and that they should use the techniques for representation and

formalization that exist. A second point is that social systems are cybernetic structures. Their components interact in complex ways through reciprocal causation and feedback loops. Every sociologist knows this. The study of the trajectory of such a system is only possible if we formalize our knowledge and assumptions about the nature of the components and their interactions, and then implement the system as a program. Sociologists have been too modest in their efforts to understand the detailed dynamics of social systems. In the natural sciences it is taken for granted that the time properties of physical, chemical and biological systems, i.e. their possible trajectories (given information about starting states), as well as their conditions of equilibrium, stability and instability, are central topics of investigation. In the sociological literature, by contrast, one rarely finds any examples of dynamic analysis of systems beyond the small group level of complexity.

4. CONCLUSION

I have tried to give some illustrations of how sociological theorizing could fruitfully draw on the advances that have taken place in Artificial Intelligence and cognitive science over the last two decades. My argument is that sociologists have a great deal to learn from these disciplines, and that the adoption of concepts, methods and tools from them would change sociologists' working habits more than their substantive concerns. I believe that they would be able to pursue traditional sociological theory better. Symbolic interactionism and parsonian functionalism are examples of traditional sociological orientations that stress actors' cognitions. It would be unfortunate if sociologists were to fail to avail themselves of the exact methods for describing and representing cognitive structure that have finally become available.

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